

Linear Motion · Practice Problem Set 1 (Solution)

Problem 1.

Given: initial velocity $v_0 = 0$ m/s, acceleration $a = 2.0$ m/s², time $t = 5.0$ s.

(a) $v = v_0 + a \cdot t = 0 + (2.0)(5.0) = 10.0$ m/s.

(b) $x = v_0 \cdot t + (1/2) \cdot a \cdot t^2 = 0 + 0.5 \cdot (2.0) \cdot (5.0)^2 = 25.0$ m.

The cyclist reaches 10.0 m/s and travels 25.0 m.